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Audit Session

ID: E23F25

- OVERVIEW
- METRICS
- FEATURE IMPORTANCE
- GEMINI REPORT
- REMEDIATION

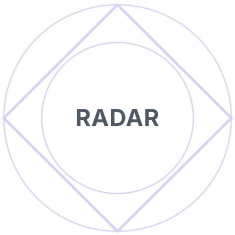
RISK ASSESSMENT

MEDIUM SEVERITY

0 of 5 metrics failed.

Audit Summary

0 of 5 core fairness metrics failed. race and gender and age are significant predictors of outcome.



FAIRNESS METRICS APR 26, 2026

Demographic Parity Violation

PASS

Measures difference in selection rates across demographic groups.

0.017 vs < 0.1 target

Caucasian AfricanAmerican Other

Equalized Odds Violation

PASS

Measures difference in true positive and false positive rates.

0.(

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ID: E23F25

Predictive Parity Violation

PASS

Measures difference in precision across groups.

0.000 vs < 0.1 target

Caucasian AfricanAmerican Other

Disparate Impact

PASS

Ratio of selection rates (min/max). Below 0.8 typically signals disparate impact.

0.853 vs > 0.8 target

Caucasian AfricanAmerican Other

Individual Fairness Violation

AMBER

Proxy based on k-NN consistency. Checks if similar inputs receive similar predictions.

0.159 vs < 0.1 target

Caucasian AfricanAmerican Other

Outcome Rate by Protected Group

Outcome Rate by Group

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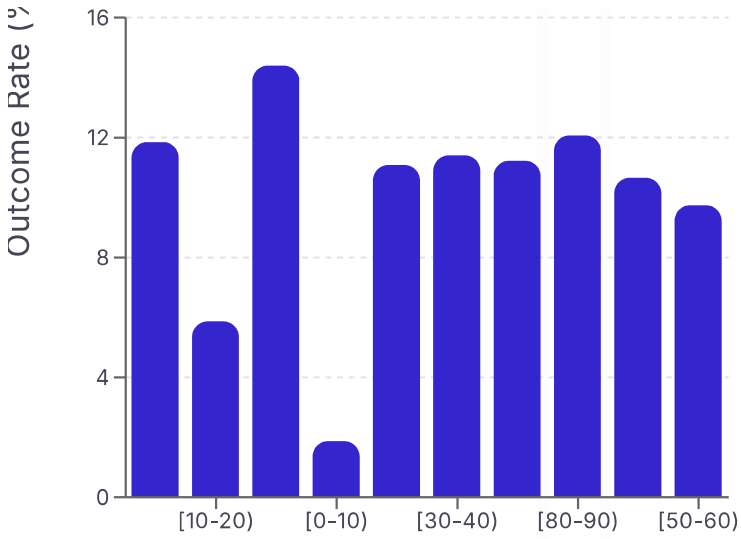
ID: E23F25



Positive outcome rates per group in the age protected attribute

What's Driving the Model?

READMITTED	+4.34
PATIENT_NBR	+0.00
ENCOUNTER_ID	+0.00
DIAG_2	+0.00
DIAG_1	+0.00
NUM_MEDICATIONS	+0.00
NUM_LAB_PROCEDURES	+0.00
INSULIN	+0.00

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Audit Session

ID: E23F25

REPORT

Audit Findings

SUMMARY

Our goal was to build a model to predict which patients are likely to be readmitted to the hospital within 30 days, ensuring it's fair to everyone. Our audit found that while the model doesn't seem to disadvantage large demographic groups (e.g., based on race or gender), it has a fundamental flaw that makes it unusable. The model is essentially 'cheating.' It's using information to make predictions that it would not have in a real-world situation—specifically, it appears to be using the answer (whether a patient was readmitted) to predict the answer. This is a critical error called 'target leakage.' Because of this, we also found an 'individual fairness' problem. This means two patients with very similar medical histories could receive different predictions from the model, which is inconsistent and unreliable. The model isn't learning the real medical reasons for readmission; it's just relying on its 'cheat sheet.' Therefore, we must not use this model. The correct action is to fix the data by removing the 'cheating' information and build a new model from the ground up. Only after this is done and we re-run our audit can we be confident that the model is both accurate and fair.

WHO IS AFFECTED

While group-level metrics do not flag specific demographic groups, the

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Audit Session

ID: E23F25

unreliable predictions.

ROOT CAUSE

The primary root cause is target leakage. The SHAP analysis shows the feature 'readmitted' has a feature importance of 4.3371, while clinically relevant features like 'diag_1' and 'diag_2' have an importance of 0.0000. This indicates the model is using the target variable (or a direct proxy) as a predictive feature, making it useless for real-world prediction and invalidating all performance and fairness metrics. The high Individual Fairness Violation is a symptom of the model's reliance on this single leaky feature rather than learning genuine patterns from patient data.

SUGGESTED:

How can I mitigate race proxy bias?

Which group is most affected by this model?

What are the top remediation steps I should take?

Ask Gemini about this au

STRATEGIC REMEDIATION PROTOCOL

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ID: E23F25

flawed and untrustworthy.

02 Conduct a thorough audit of the entire data pipeline to identify and remove the 'readmitted' feature and any other features that leak information about the target variable from the training dataset.

03 Retrain the model from scratch on the corrected dataset. After retraining, generate new SHAP values to ensure the model is leveraging clinically relevant features.

04 Perform a complete re-evaluation of the new model's fairness, paying close attention to both group and individual fairness metrics to ensure the retrained model does not introduce new biases.

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Audit Session

ID: E23F25

'AICresult' to understand if missingness is correlated with protected attributes and apply an appropriate imputation strategy to avoid introducing bias.

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